

Shai Cahana

054-784-1589 | shaicahana@gmail.com | [LinkedIn](#) | <http://shaicahana.com/>

Professional Summary

Applied Physics M.Sc. graduate with hands-on experience in clinical data analysis, statistical modeling, computer vision, and machine learning. Currently working as a Data Analyst & Research Coordinator at Maccabi Dent, where my work contributed to a peer-reviewed publication in a Wiley journal. Seeking to leverage a strong technical foundation in data science and algorithm development to drive impactful research and product outcomes.

Technical Skills

- **Programming:** Python, MATLAB, SQL
- **Image Processing & Computer Vision:** OpenCV, scikit-image
- **Data Processing:** pandas, SciPy, NumPy, PyTorch, TensorFlow, seaborn, CVAT
- **Technologies:** OOP, Git, Linux, AWS, Blender
- **Data Analysis:** Statistical analysis, survival analysis (Kaplan–Meier, Cox regression), exploratory data analysis

Professional Experience

Data Analyst & Research Coordinator – Maccabi Dent (May 2025 – Present)

- Extracted and processed large-scale clinical data from electronic health records, building SQL pipelines and performing statistical analysis in Python to support research conclusions.
- Managed ethics committee approvals and data security protocols in compliance with institutional standards.
- Applied survival analysis methods (Kaplan–Meier, Cox proportional hazards) and GEE models to evaluate clinical outcomes across cohorts of tens of thousands of patients.
- Supported the Quality Control division with data extractions and audits of medical billing records to identify discrepancies and ensure billing accuracy.

Data Scientist – Deep Vision Systems (Jan 2025 – Mar 2025)

- Curated benchmark datasets and validated model performance for production integration.
- Designed synthetic data scenes in Blender for model training and validation.
- Analyzed user behavior data to refine app performance and engagement metrics.

Security Engineer – Wireless Connectivity Solutions, Intel (2017 – 2018)

- Developed Python scripts for automating data extraction and analysis tasks
- Implemented data analysis tools for large-scale security log processing

OSINT Researcher – Special Operations, Intelligence Force, IDF (2013 – 2015)

- Performed research across multiple intelligence sources to extract and synthesize actionable findings.
- Created reports and presentations to communicate findings to decision-makers

Education

AI and Deep Learning Development – Google Reichman Tech School (2024)

- Advanced course covering AI algorithms and techniques, with practical applications in data analysis and machine learning.

M.Sc. Applied Physics - The Hebrew University of Jerusalem (2020 - 2024)

- 93 GPA, awarded dean's list. Thesis: Morphogenesis of leaf tissue.
 - Developed computer vision algorithms for feature detection in biological systems
 - Implemented data processing pipelines for large-scale experimental datasets

B.Sc. Electrical & Computer Engineering - The Hebrew University of Jerusalem (2015 - 2019)

- Relevant courses: Intro to CS, Data Structures, Algorithms, Computer Vision

Projects

4D Growth Tracker

- 4D Growth Tracker: built a 3D timelapse setup used to analyze flower opening dynamics over time, integrating computer vision techniques with 3D-printed hardware, Arduino & cameras. The scanner worked continuously, generating "4D" scans of the plants.

Publications

- Haim D., Eskander-Hashoul L., Better H., Meinster I., **Cahana S.**, Chaushu G., Asbi T. (2026). Survival and Risk Factors of Dental Implants in the Anterior Mandible: A Cohort Study. *Clinical Implant Dentistry and Related Research*. <https://doi.org/10.1111/cid.70142>